BITCOIN MARKET SENTIMENT VS TRADER PERFORMANCE ANALYSIS

1. Objective

This analysis aims to examine the correlation between Bitcoin market sentiment and trader actions based on the Fear & Greed Index dataset and Hyperliquid historical trading data. The analysis is designed to determine the impact of various market emotions (Fear, Greed, Extreme Fear, and Extreme Greed) on trading activity, profitability, and trader decision-making.

2. Datasets Used

Dataset 1 — Bitcoin Fear & Greed Index

Columns used: Date Classification

The dataset classifies market sentiment into:

Extreme Fear

Fear

Neutral

Greed

Extreme Greed

This dataset contains historical trader data that is hyperliquid.This dataset includes hyperliquid historical trader data.

Columns used: account coin

execution_price

size_tokens

size_usd

side

closed_pnl

fee

timestamp

3. Data Preprocessing

The following pre-processing steps were carried out:

Converted timestamps to date format

Standardized column names

Removed missing values

Combined both data sets by date

Developed new metrics including:

trade_value

ROI

profit/loss indicator

Additional calculated features:

is_profit

roi

trade_value

4. Exploratory Data Analysis

The analysis was based on:

Trade distribution across sentiments

Profitability trends

ROI behavior

Win rates

Trading activity patterns

Trade size behavior

To detect patterns in trader activity across various market sentiment scenarios, several visualizations were developed using Matplotlib and Seaborn.

5. Key Findings

5.1 Trade Activity Was Highest During Fear and Greed Phases . The analysis of the number of trades revealed that the most trades were made when the market was in Fear and Greed.

Sentiment	Trade Count
Fear	61,837
Greed	50,030
Extreme Greed	39,992
Neutral	37,686
Extreme Fear	21,400

This means that traders are much more active during emotionally charged market times.

The highest average profitability was generated by 5.2 Extreme Greed.

Average closed PnL analysis showed:

Sentiment	Average Closed PnL
Extreme Greed	67.89
Fear	54.29
Greed	42.74
Extreme Fear	34.54
Neutral	34.31

The average profits were highest for Extreme Greed, indicating that when bulls are strong, there are greater profit opportunities for active traders.

The 5.3 Win Rate was the highest during Extreme Greed.

Sentiment	Win Rate (%)
Extreme Greed	46.49%
Fear	42.08%
Neutral	39.70%
Greed	38.48%
Extreme Fear	37.06%

This indicates that traders had more success in trading when the market was in a very bullish state.

5.4 ROI Analysis showed better performance in Greed Phases.

Sentiment	Average ROI
Extreme Greed	0.0400
Greed	0.0196
Fear	0.0154
Neutral	0.0099
Extreme Fear	0.0043

The results of the ROI analysis indicated that market optimism had a positive effect on trader returns. Extreme Greed was significantly higher than all other sentiment categories.

6. Behavioral Insights

The analysis indicates that market sentiment has a significant impact on trader behavior and profitability.

Key behavioral observations:

Traders are more active during Fear and Greed periods.

Extreme Greed had the highest profitability and highest win rate.

Volatility continued to draw trading activity, as fear markets remained active.

Neutral markets generated lower profitability and lower engagement

Trader participation and opportunity generation seem to be greater during emotional market conditions.

Conclusion

This analysis shows that there is a clear correlation between Bitcoin market sentiment and trader performance.

The results indicate that:

Profitability and ROI are higher in markets driven by greed.

Trader participation rises during emotionally charged times.

Market sentiment can serve as a valuable behavioral indicator for understanding trading activity

In conclusion, sentiment analysis can offer valuable insights for crafting more intelligent trading strategies and making better market timing decisions.